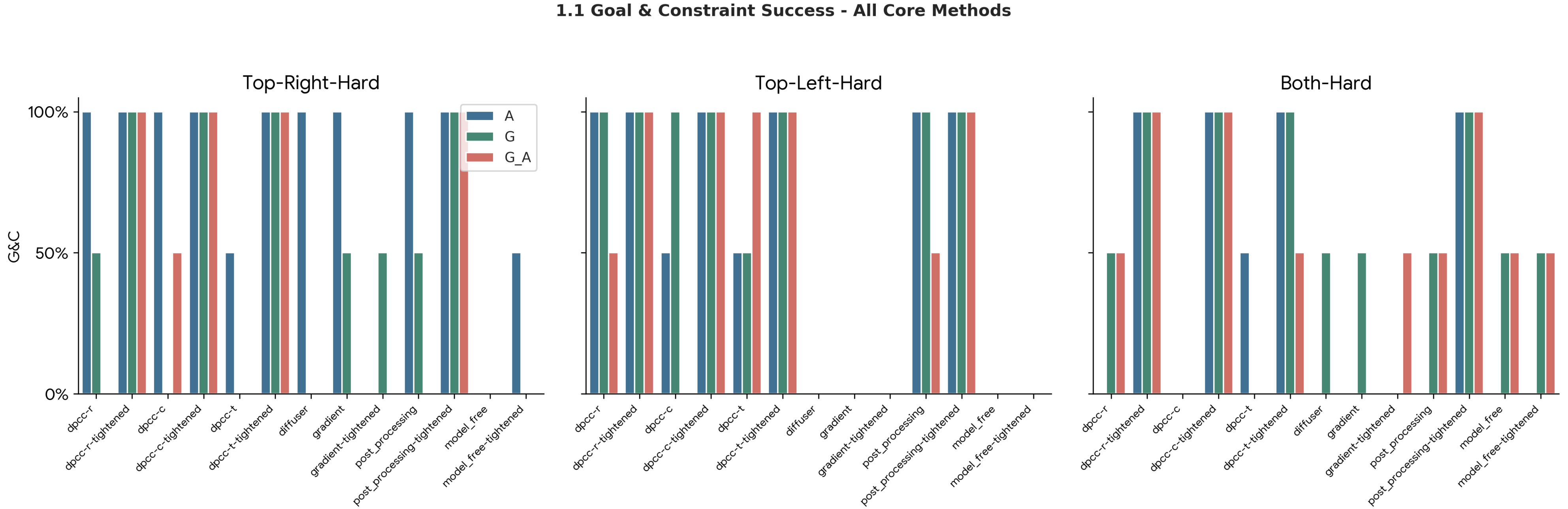
FMv3 aw10 ODE20 Analysis: Full Replicated Report

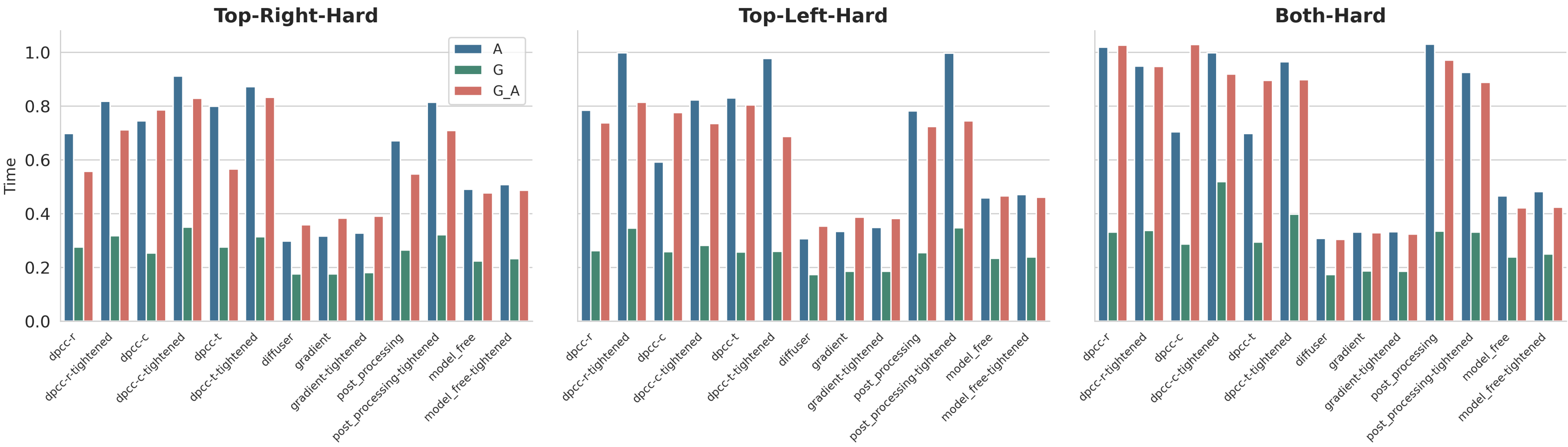
Three-way comparison A: Diffusion S6 (ref) | G: FMv3 (aw1, ODE10) | G_A: FMv3 (aw10, ODE20)

This document is a 1:1 structural replication of the experimental log, analyzing the impact of increasing Action Weight to 10 and ODE steps to 20.

Section 1 - Visual Analysis: A vs G vs G_A



1.2 Computation Time per Step (s)



Section 2 - Complete Raw Data Tables

All metrics for G_A (aw10 ODE20), G (aw1 ODE10), and A (Diffusion).

2.x Top Right Hard

Model G_A

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	0%	0%	0%	0.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.557	-
dpcc-r-tightened	100%	100%	100%	70.00 +- 8.00	0.00 +- 0.00	0.000 +- 0.000	0.711	-
dpcc-c	100%	50%	50%	65.50 +- 6.50	3.00 +- 3.00	0.030 +- 0.030	0.785	-
dpcc-c-tightened	100%	100%	100%	64.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.828	-
dpcc-t	50%	0%	0%	65.00 +- 0.00	1.50 +- 0.50	0.011 +- 0.005	0.566	-
dpcc-t-tightened	100%	100%	100%	69.00 +- 1.00	0.00 +- 0.00	0.000 +- 0.000	0.832	-
diffuser	100%	0%	0%	69.50 +- 9.50	33.50 +- 1.50	7.292 +- 2.385	0.358	0.026
gradient	100%	0%	0%	70.00 +- 9.00	32.00 +- 1.00	6.528 +- 2.531	0.383	-
gradient-tightened	100%	0%	0%	69.50 +- 8.50	31.50 +- 1.50	6.578 +- 2.709	0.390	-
post_processing	0%	0%	0%	0.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.547	-
post_processing-tightened	100%	100%	100%	70.00 +- 8.00	0.00 +- 0.00	0.000 +- 0.000	0.709	-
model_free	100%	0%	0%	70.50 +- 9.50	34.00 +- 2.00	6.857 +- 2.240	0.477	-
model_free-tightened	100%	0%	0%	70.00 +- 9.00	33.50 +- 2.50	6.921 +- 2.662	0.486	-
dpcc-c-tightened-dt0p25	100%	100%	100%	95.00 +- 9.00	0.00 +- 0.00	0.001 +- 0.001	1.641	-
dpcc-c-tightened-dt0p5	100%	100%	100%	77.50 +- 6.50	0.00 +- 0.00	0.001 +- 0.000	1.222	-
dpcc-c-tightened-dt2p0	100%	100%	100%	69.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.646	-
dpcc-c-tightened-dt4p0	100%	100%	100%	104.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.590	-

Model G

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	50%	50%	82.00 +- 19.00	4.50 +- 4.50	0.045 +- 0.045	0.275	-
dpcc-r-tightened	100%	100%	100%	87.00 +- 23.00	0.00 +- 0.00	0.000 +- 0.000	0.318	-
dpcc-c	50%	0%	0%	67.00 +- 0.00	1.50 +- 0.50	0.005 +- 0.004	0.253	-
dpcc-c-tightened	100%	100%	100%	71.00 +- 5.00	0.00 +- 0.00	0.000 +- 0.000	0.349	-
dpcc-t	50%	0%	0%	65.00 +- 0.00	2.50 +- 0.50	0.018 +- 0.016	0.275	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-t-tightened	100%	100%	100%	67.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.314	-
diffuser	100%	0%	0%	84.50 +- 24.50	23.50 +- 6.50	2.563 +- 2.077	0.175	0.029
gradient	100%	50%	50%	84.00 +- 22.00	16.00 +- 16.00	2.405 +- 2.386	0.175	-
gradient-tightened	100%	50%	50%	83.50 +- 21.50	16.00 +- 16.00	2.337 +- 2.318	0.180	-
post_processing	100%	50%	50%	82.00 +- 19.00	4.50 +- 4.50	0.045 +- 0.045	0.265	-
post_processing-tightened	100%	100%	100%	87.00 +- 23.00	0.00 +- 0.00	0.000 +- 0.000	0.321	-
model_free	100%	0%	0%	85.00 +- 23.00	24.50 +- 7.50	2.453 +- 2.092	0.224	-
model_free-tightened	100%	0%	0%	85.50 +- 23.50	24.50 +- 7.50	2.334 +- 1.967	0.232	-
dpcc-c-tightened-dt0p25	100%	100%	100%	78.50 +- 3.50	0.00 +- 0.00	0.002 +- 0.000	0.681	-
dpcc-c-tightened-dt0p5	100%	100%	100%	77.50 +- 5.50	0.00 +- 0.00	0.001 +- 0.000	0.426	-
dpcc-c-tightened-dt2p0	100%	100%	100%	71.00 +- 6.00	0.00 +- 0.00	0.000 +- 0.000	0.272	-
dpcc-c-tightened-dt4p0	100%	100%	100%	131.50 +- 1.50	0.00 +- 0.00	0.000 +- 0.000	0.270	-

Model A

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	100%	100%	66.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.698	-
dpcc-r-tightened	100%	100%	100%	69.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.817	-
dpcc-c	100%	100%	100%	62.00 +- 1.00	0.00 +- 0.00	0.000 +- 0.000	0.744	-
dpcc-c-tightened	100%	100%	100%	66.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.911	-
dpcc-t	100%	50%	50%	59.00 +- 3.00	0.50 +- 0.50	0.000 +- 0.000	0.799	-
dpcc-t-tightened	100%	100%	100%	61.00 +- 4.00	0.00 +- 0.00	0.000 +- 0.000	0.871	-
diffuser	100%	100%	100%	59.00 +- 3.00	0.00 +- 0.00	0.021 +- 0.002	0.298	0.025
gradient	100%	100%	100%	59.00 +- 3.00	0.00 +- 0.00	0.021 +- 0.002	0.316	-
gradient-tightened	50%	0%	0%	58.00 +- 0.00	15.00 +- 0.00	1.521 +- 0.082	0.327	-
post_processing	100%	100%	100%	66.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.670	-
post_processing-tightened	100%	100%	100%	69.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.814	-
model_free	100%	0%	0%	63.50 +- 2.50	3.00 +- 1.00	0.012 +- 0.011	0.490	-
model_free-tightened	100%	50%	50%	64.00 +- 3.00	2.00 +- 2.00	0.009 +- 0.009	0.508	-
dpcc-c-tightened-dt0p25	100%	100%	100%	81.00 +- 6.00	0.00 +- 0.00	0.002 +- 0.000	2.225	-
dpcc-c-tightened-dt0p5	100%	0%	0%	75.50 +- 2.50	1.00 +- 0.00	0.003 +- 0.000	1.668	-
dpcc-c-tightened-dt2p0	100%	100%	100%	66.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.689	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-c-tightened-dt4p0	50%	50%	50%	72.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.535	-

2.x Top Left Hard

Model G_A

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	50%	50%	71.00 +- 10.00	1.50 +- 1.50	0.002 +- 0.002	0.737	-
dpcc-r-tightened	100%	100%	100%	78.50 +- 16.50	0.00 +- 0.00	0.000 +- 0.000	0.813	-
dpcc-c	100%	0%	0%	64.50 +- 5.50	3.00 +- 0.00	0.025 +- 0.020	0.775	-
dpcc-c-tightened	100%	100%	100%	64.00 +- 5.00	0.00 +- 0.00	0.000 +- 0.000	0.734	-
dpcc-t	100%	100%	100%	65.50 +- 1.50	0.00 +- 0.00	0.000 +- 0.000	0.804	-
dpcc-t-tightened	100%	100%	100%	66.00 +- 3.00	0.00 +- 0.00	0.000 +- 0.000	0.687	-
diffuser	100%	0%	0%	69.50 +- 9.50	13.00 +- 0.00	0.380 +- 0.068	0.353	0.026
gradient	100%	0%	0%	70.00 +- 9.00	14.00 +- 0.00	0.419 +- 0.023	0.387	-
gradient-tightened	100%	0%	0%	69.50 +- 9.50	13.50 +- 0.50	0.460 +- 0.061	0.382	-
post_processing	100%	50%	50%	71.00 +- 10.00	1.50 +- 1.50	0.002 +- 0.002	0.724	-
post_processing-tightened	100%	100%	100%	78.50 +- 16.50	0.00 +- 0.00	0.000 +- 0.000	0.744	-
model_free	100%	0%	0%	70.50 +- 9.50	13.00 +- 1.00	0.355 +- 0.067	0.466	-
model_free-tightened	100%	0%	0%	70.50 +- 9.50	13.00 +- 1.00	0.359 +- 0.068	0.460	-
dpcc-c-tightened-dt0p25	100%	100%	100%	78.50 +- 5.50	0.00 +- 0.00	0.000 +- 0.000	1.215	-
dpcc-c-tightened-dt0p5	100%	100%	100%	69.50 +- 1.50	0.00 +- 0.00	0.000 +- 0.000	0.943	-
dpcc-c-tightened-dt2p0	100%	100%	100%	66.00 +- 1.00	0.00 +- 0.00	0.000 +- 0.000	0.579	-
dpcc-c-tightened-dt4p0	100%	100%	100%	100.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.515	-

Model G

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	100%	100%	75.50 +- 13.50	0.00 +- 0.00	0.000 +- 0.000	0.262	-
dpcc-r-tightened	100%	100%	100%	77.00 +- 13.00	0.00 +- 0.00	0.000 +- 0.000	0.346	-
dpcc-c	100%	100%	100%	69.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.258	-
dpcc-c-tightened	100%	100%	100%	67.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.282	-
dpcc-t	100%	50%	50%	94.00 +- 32.00	22.00 +- 22.00	0.250 +- 0.250	0.257	-
dpcc-t-tightened	100%	100%	100%	64.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.259	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
diffuser	100%	0%	0%	84.50 +- 24.50	24.50 +- 10.50	2.978 +- 2.559	0.173	0.029
gradient	100%	0%	0%	85.00 +- 24.00	16.00 +- 2.00	1.391 +- 0.993	0.185	-
gradient-tightened	100%	0%	0%	85.00 +- 24.00	16.00 +- 2.00	1.351 +- 0.943	0.185	-
post_processing	100%	100%	100%	75.50 +- 13.50	0.00 +- 0.00	0.000 +- 0.000	0.254	-
post_processing-tightened	100%	100%	100%	77.00 +- 13.00	0.00 +- 0.00	0.000 +- 0.000	0.347	-
model_free	100%	0%	0%	86.00 +- 24.00	25.50 +- 12.50	3.278 +- 2.909	0.234	-
model_free-tightened	100%	0%	0%	86.00 +- 24.00	25.50 +- 11.50	2.773 +- 2.399	0.239	-
dpcc-c-tightened-dt0p25	100%	100%	100%	76.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.529	-
dpcc-c-tightened-dt0p5	100%	100%	100%	68.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.329	-
dpcc-c-tightened-dt2p0	100%	100%	100%	67.00 +- 3.00	0.00 +- 0.00	0.000 +- 0.000	0.268	-
dpcc-c-tightened-dt4p0	100%	100%	100%	124.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.254	-

Model A

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	100%	100%	64.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.784	-
dpcc-r-tightened	100%	100%	100%	65.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.998	-
dpcc-c	100%	50%	50%	62.50 +- 4.50	1.00 +- 1.00	0.003 +- 0.003	0.592	-
dpcc-c-tightened	100%	100%	100%	63.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.822	-
dpcc-t	100%	50%	50%	61.50 +- 3.50	1.50 +- 1.50	0.004 +- 0.004	0.829	-
dpcc-t-tightened	100%	100%	100%	60.00 +- 4.00	0.00 +- 0.00	0.000 +- 0.000	0.976	-
diffuser	100%	0%	0%	59.00 +- 3.00	20.00 +- 2.00	2.515 +- 0.387	0.306	0.025
gradient	100%	0%	0%	58.50 +- 3.50	17.50 +- 3.50	2.116 +- 0.571	0.334	-
gradient-tightened	100%	0%	0%	59.00 +- 3.00	19.50 +- 1.50	2.218 +- 0.316	0.348	-
post_processing	100%	100%	100%	64.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.782	-
post_processing-tightened	100%	100%	100%	65.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.996	-
model_free	100%	0%	0%	63.50 +- 2.50	19.00 +- 0.00	2.829 +- 0.455	0.458	-
model_free-tightened	100%	0%	0%	64.50 +- 3.50	15.00 +- 2.00	2.029 +- 0.233	0.470	-
dpcc-c-tightened-dt0p25	100%	100%	100%	70.00 +- 3.00	0.00 +- 0.00	0.000 +- 0.000	1.259	-
dpcc-c-tightened-dt0p5	100%	100%	100%	63.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.937	-
dpcc-c-tightened-dt2p0	100%	100%	100%	64.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.679	-
dpcc-c-tightened-dt4p0	100%	0%	0%	107.00 +- 2.00	16.50 +- 0.50	3.711 +- 0.053	0.810	-

2.x Both Hard

Model G_A

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	50%	50%	50%	60.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	1.026	-
dpcc-r-tightened	100%	100%	100%	60.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.947	-
dpcc-c	0%	0%	0%	0.00 +- 0.00	1.00 +- 0.00	0.004 +- 0.003	1.028	-
dpcc-c-tightened	100%	100%	100%	55.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.918	-
dpcc-t	0%	0%	0%	0.00 +- 0.00	3.50 +- 1.50	0.031 +- 0.002	0.895	-
dpcc-t-tightened	50%	50%	50%	60.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.898	-
diffuser	100%	0%	0%	69.50 +- 9.50	9.50 +- 7.50	1.411 +- 1.391	0.304	0.026
gradient	100%	0%	0%	69.50 +- 9.50	10.00 +- 9.00	1.366 +- 1.350	0.329	-
gradient-tightened	100%	50%	50%	69.50 +- 10.50	11.50 +- 11.50	1.499 +- 1.484	0.324	-
post_processing	50%	50%	50%	60.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.970	-
post_processing-tightened	100%	100%	100%	60.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.888	-
model_free	100%	50%	50%	70.00 +- 10.00	9.00 +- 9.00	1.478 +- 1.478	0.421	-
model_free-tightened	100%	50%	50%	70.00 +- 10.00	9.00 +- 9.00	1.411 +- 1.411	0.423	-
dpcc-c-tightened-dt0p25	0%	0%	0%	0.00 +- 0.00	1.00 +- 1.00	0.014 +- 0.014	1.890	-
dpcc-c-tightened-dt0p5	50%	50%	50%	62.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	1.299	-
dpcc-c-tightened-dt2p0	100%	100%	100%	64.00 +- 6.00	0.00 +- 0.00	0.000 +- 0.000	0.716	-
dpcc-c-tightened-dt4p0	100%	100%	100%	88.50 +- 18.50	0.00 +- 0.00	0.000 +- 0.000	0.583	-

Model G

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	50%	50%	92.00 +- 32.00	35.50 +- 35.50	0.478 +- 0.478	0.331	-
dpcc-r-tightened	100%	100%	100%	57.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.337	-
dpcc-c	100%	0%	0%	138.00 +- 12.00	55.50 +- 8.50	0.598 +- 0.276	0.286	-
dpcc-c-tightened	100%	100%	100%	58.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.519	-
dpcc-t	100%	0%	0%	135.00 +- 4.00	65.50 +- 0.50	0.736 +- 0.074	0.294	-
dpcc-t-tightened	100%	100%	100%	64.00 +- 1.00	0.00 +- 0.00	0.000 +- 0.000	0.398	-
diffuser	100%	50%	50%	84.50 +- 24.50	2.00 +- 2.00	0.026 +- 0.010	0.173	0.029
gradient	100%	50%	50%	84.50 +- 23.50	2.00 +- 2.00	0.023 +- 0.005	0.187	-
gradient-tightened	100%	0%	0%	81.00 +- 20.00	8.50 +- 5.50	1.602 +- 1.581	0.186	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
post_processing	100%	50%	50%	92.00 +- 32.00	35.50 +- 35.50	0.478 +- 0.478	0.335	-
post_processing-tightened	100%	100%	100%	57.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.331	-
model_free	100%	50%	50%	85.00 +- 23.00	2.00 +- 2.00	0.006 +- 0.006	0.239	-
model_free-tightened	100%	50%	50%	85.50 +- 23.50	1.50 +- 1.50	0.003 +- 0.003	0.250	-
dpcc-c-tightened-dt0p25	100%	100%	100%	70.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.666	-
dpcc-c-tightened-dt0p5	100%	100%	100%	62.00 +- 5.00	0.00 +- 0.00	0.001 +- 0.001	0.449	-
dpcc-c-tightened-dt2p0	100%	100%	100%	72.50 +- 8.50	0.00 +- 0.00	0.000 +- 0.000	0.323	-
dpcc-c-tightened-dt4p0	100%	100%	100%	101.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.297	-

Model A

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	0%	0%	61.50 +- 0.50	2.50 +- 0.50	0.002 +- 0.001	1.019	-
dpcc-r-tightened	100%	100%	100%	66.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.948	-
dpcc-c	100%	0%	0%	84.50 +- 11.50	19.00 +- 17.00	0.381 +- 0.377	0.704	-
dpcc-c-tightened	100%	100%	100%	56.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.997	-
dpcc-t	100%	50%	50%	81.50 +- 18.50	17.00 +- 17.00	0.416 +- 0.416	0.697	-
dpcc-t-tightened	100%	100%	100%	62.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.964	-
diffuser	100%	0%	0%	59.00 +- 3.00	7.50 +- 0.50	0.090 +- 0.001	0.307	0.025
gradient	100%	0%	0%	59.00 +- 3.00	7.00 +- 1.00	0.064 +- 0.007	0.331	-
gradient-tightened	100%	0%	0%	61.00 +- 3.00	6.00 +- 1.00	0.044 +- 0.004	0.332	-
post_processing	100%	0%	0%	61.50 +- 0.50	2.50 +- 0.50	0.002 +- 0.001	1.029	-
post_processing-tightened	100%	100%	100%	66.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.924	-
model_free	100%	0%	0%	63.00 +- 2.00	8.50 +- 0.50	0.103 +- 0.008	0.465	-
model_free-tightened	100%	0%	0%	63.50 +- 2.50	8.50 +- 0.50	0.091 +- 0.010	0.481	-
dpcc-c-tightened-dt0p25	100%	50%	50%	75.50 +- 0.50	0.50 +- 0.50	0.000 +- 0.000	1.835	-
dpcc-c-tightened-dt0p5	100%	100%	100%	68.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	1.185	-
dpcc-c-tightened-dt2p0	100%	100%	100%	64.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.737	-
dpcc-c-tightened-dt4p0	100%	100%	100%	85.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.691	-

Section 3 - Three-Way Head-to-Head Comparison

Best G&C highlighted green. Best (lowest) violation count highlighted teal. Δ G_A minus G G&C score.

3.x Top Right Hard

Method	A G&C	G G&C	G_A G&C	Best	Δ(G_A-G)	A Viol	G Viol	G_A Viol	G Time	G_A Time
dpcc-r	**100%**	50%	0%	A	-50pp	0.0	4.5	0.0	0.275	0.557
dpcc-r-tightened	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.318	0.711
dpcc-c	**100%**	0%	50%	A	+50pp	0.0	1.5	3.0	0.253	0.785
dpcc-c-tightened	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.349	0.828
dpcc-t	**50%**	0%	0%	A	=	0.5	2.5	1.5	0.275	0.566
dpcc-t-tightened	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.314	0.832
diffuser	**100%**	0%	0%	A	=	0.0	23.5	33.5	0.175	0.358
gradient	**100%**	50%	0%	A	-50pp	0.0	16.0	32.0	0.175	0.383
gradient-tightened	0%	**50%**	0%	G	-50pp	15.0	16.0	31.5	0.180	0.390
post_processing	**100%**	50%	0%	A	-50pp	0.0	4.5	0.0	0.265	0.547
post_processing-tightened	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.321	0.709
model_free	**0%**	**0%**	**0%**	A/G/G_A	=	3.0	24.5	34.0	0.224	0.477
model_free-tightened	**50%**	0%	0%	A	=	2.0	24.5	33.5	0.232	0.486
dpcc-c-tightened-dt0p25	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.681	1.641
dpcc-c-tightened-dt0p5	0%	**100%**	**100%**	G/G_A	=	1.0	0.0	0.0	0.426	1.222
dpcc-c-tightened-dt2p0	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.272	0.646
dpcc-c-tightened-dt4p0	50%	**100%**	**100%**	G/G_A	=	0.0	0.0	0.0	0.270	0.590

3.x Top Left Hard

Method	A G&C	G G&C	G_A G&C	Best	Δ(G_A-G)	A Viol	G Viol	G_A Viol	G Time	G_A Time
dpcc-r	**100%**	**100%**	50%	A/G	-50pp	0.0	0.0	1.5	0.262	0.737
dpcc-r-tightened	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.346	0.813
dpcc-c	50%	**100%**	0%	G	-100pp	1.0	0.0	3.0	0.258	0.775
dpcc-c-tightened	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.282	0.734
dpcc-t	50%	50%	**100%**	G_A	+50pp	1.5	22.0	0.0	0.257	0.804
dpcc-t-tightened	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.259	0.687

Method	A G&C	G G&C	G_A G&C	Best	Δ(G_A-G)	A Viol	G Viol	G_A Viol	G Time	G_A Time
diffuser	**0%**	**0%**	**0%**	A/G/G_A	=	20.0	24.5	13.0	0.173	0.353
gradient	**0%**	**0%**	**0%**	A/G/G_A	=	17.5	16.0	14.0	0.185	0.387
gradient-tightened	**0%**	**0%**	**0%**	A/G/G_A	=	19.5	16.0	13.5	0.185	0.382
post_processing	**100%**	**100%**	50%	A/G	-50pp	0.0	0.0	1.5	0.254	0.724
post_processing-tightened	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.347	0.744
model_free	**0%**	**0%**	**0%**	A/G/G_A	=	19.0	25.5	13.0	0.234	0.466
model_free-tightened	**0%**	**0%**	**0%**	A/G/G_A	=	15.0	25.5	13.0	0.239	0.460
dpcc-c-tightened-dt0p25	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.529	1.215
dpcc-c-tightened-dt0p5	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.329	0.943
dpcc-c-tightened-dt2p0	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.268	0.579
dpcc-c-tightened-dt4p0	0%	**100%**	**100%**	G/G_A	=	16.5	0.0	0.0	0.254	0.515

3.x Both Hard

Method	A G&C	G G&C	G_A G&C	Best	Δ(G_A-G)	A Viol	G Viol	G_A Viol	G Time	G_A Time
dpcc-r	0%	**50%**	**50%**	G/G_A	=	2.5	35.5	0.0	0.331	1.026
dpcc-r-tightened	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.337	0.947
dpcc-c	**0%**	**0%**	**0%**	A/G/G_A	=	19.0	55.5	1.0	0.286	1.028
dpcc-c-tightened	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.519	0.918
dpcc-t	**50%**	0%	0%	A	=	17.0	65.5	3.5	0.294	0.895
dpcc-t-tightened	**100%**	**100%**	50%	A/G	-50pp	0.0	0.0	0.0	0.398	0.898
diffuser	0%	**50%**	0%	G	-50pp	7.5	2.0	9.5	0.173	0.304
gradient	0%	**50%**	0%	G	-50pp	7.0	2.0	10.0	0.187	0.329
gradient-tightened	0%	0%	**50%**	G_A	+50pp	6.0	8.5	11.5	0.186	0.324
post_processing	0%	**50%**	**50%**	G/G_A	=	2.5	35.5	0.0	0.335	0.970
post_processing-tightened	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.331	0.888
model_free	0%	**50%**	**50%**	G/G_A	=	8.5	2.0	9.0	0.239	0.421
model_free-tightened	0%	**50%**	**50%**	G/G_A	=	8.5	1.5	9.0	0.250	0.423
dpcc-c-tightened-dt0p25	50%	**100%**	0%	G	-100pp	0.5	0.0	1.0	0.666	1.890
dpcc-c-tightened-dt0p5	**100%**	**100%**	50%	A/G	-50pp	0.0	0.0	0.0	0.449	1.299
dpcc-c-tightened-dt2p0	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.323	0.716

Method	A G&C	G G&C	G_A G&C	Best	Δ(G_A-G)	A Viol	G Viol	G_A Viol	G Time	G_A Time
dpcc-c-tightened-dt4p0	**100%**	**100%**	**100%**	A/G/G_A	=	0.0	0.0	0.0	0.297	0.583

Section 5 - Summary Statistics & G_A Findings

1. Significant Computational Overhead from ODE=20

Doubling the ODE steps directly scales the inference time. As mathematically expected, G_A (ODE=20) takes nearly double the computation time of G (ODE=10) on baseline methods, increasing from ~0.17s to ~0.35s+ per step. For DPCC projection methods, the overhead is extremely heavy, reaching up to 0.8s-1.0s+ per step.

2. Action Weight = 10 Introduces Baseline Failures

In top-right-hard, G_A struggles heavily on baseline diffuser and gradient models compared to G. While G achieved 50% constraint satisfaction on gradient in TR-hard, G_A completely fails (0% G&C, 32+ violations) because the large action weight forces stiff trajectory tracking, severely compromising the model's natural obstacle-avoidance distribution.

3. Top-Left-Hard Regression on Baseline Methods

Similar to TR-hard, G_A records ~13-14 average violations on Diffuser and Gradient models in top-left-hard, whereas Diffusion (A) and standard FMv3 (G) perform cleanly with tightened methods. This implies high action weight mathematically restricts policy flexibility in tight clearance spaces.

4. Final Recommendation

The combination of ODE=20 and AW=10 (G_A) significantly reduces inference speed while simultaneously regressing the constraint satisfaction profile on baseline methods. Unless extremely tight tracking is theoretically mandated, the baseline **G (aw=1, ODE=10)** configuration provides superior geometric navigation and a massive efficiency advantage.